

Quiz - 2

1) Complement :

$$AB (BC' + (C'D') \cdot (A' + C))$$

dual form :

$$(A + B) + ((B + C') \cdot (C' + D') + (A' \cdot C))$$

Complement form :

$$(A' + B') + ((B' + C) \cdot (C + D) + (AC'))$$

11)

A	B	C	(A+B)	(A+C)	(A+B)(A+C)	BC	A+BC
0	0	0	0	0	0	0	0
0	0	1	0	1	0	0	0
0	1	0	1	0	0	0	0
0	1	1	1	1	1	1	1
1	0	0	1	1	1	0	1
1	0	1	1	1	1	0	1

1	1	0	1	1	1	0	1
1	1	1	1	1	1	1	1

$$\therefore \text{LHS} = \text{RHS}$$

$$\text{iii) } F(A, B, C) = \sum (0, 4, 6)$$

$$= \sum (000, 100, 110)$$

$$= A'B'C' + AB'C' + ABC'$$

$$= B'C'(A' + A) + ABC'$$

$$= B'C' + ABC'$$

$$= C'(B' + AB)$$

$$= C'(B' + A)(B' + B)$$

$$= C'(A + B')$$

$$\text{iv) } F(A, B, C) = A + B'C$$

$$= (A + B')(A + C)$$

$$= (A + B' + CC') (A + C + BB')$$

$$= (A + B' + C) (A + B' + C') (A + B + C) \underline{(A + B' + C)}$$

$$= \pi (010, 011, 000)$$

$$= \pi (0, 2, 3)$$